

- 8 (a) (i)** A region of space where a force is experienced and is represented by lines of forces. B1

*Comments: Some candidates are not awarded marks if their answers are specific to “charge” or “mass” but question asked for force-field in general.*

- (ii)** Field strength was related to the closeness of the lines of force. B1  
The field strength is greater if the lines of force are closer to each other. B1

*Comments: This part is well done.*

- (b) (i)** Mass B1  
**(ii)** Positive charge B1  
**(iii)** North Pole B1

*Comments: Majority of the candidate did not read the question carefully and did not realise that the force must be in the same direction as the field as stated in the question.*

- (c) (i)** Magnetic force provides centripetal force  
 $Bqv = mv^2 / r$   
 $r = (1.67 \times 10^{-27} \times 4.5 \times 10^6) / (1.2 \times 1.6 \times 10^{-19})$  C1  
 $r = 0.039 \text{ m}$   
 $= 3.9 \text{ cm}$  A1

*Comments: This part is well done*

- (ii)** circular path with the correct radius C1  
clockwise C1  
exit field at an angle below horizontal and pointing to the right C1

straight line tangential outside field C1

*Comments: Candidates need to use compass to draw circle and ruler to draw straight line to earn the full credit.*

(d) (i) arrow pointing up the page B1

*Comments: Candidates need to use ruler to draw straight line to earn the full credit.*

(ii)  $Bqv = Eq$  C1

$$E = 1.2 \times 4.5 \times 10^6$$

$$= 5.4 \times 10^6 \text{ V m}^{-1} \quad \text{A1}$$

*Comments: A small number of candidates used  $E = Q/(4\pi\epsilon_0 r^2)$  which is only applicable if field is produced by a point charge.*

(e) gravitational force very small compared with magnetic force  
and/or electric force B1

*Comments: Must do comparison. Also must compare quantities with same unit, e.g. compare gravitational force with electric force rather than compared mass with charge.*

(f) (i) magnetic force > electric force M1

So deflects upwards A1

(ii)  $Bqv = qE$ , forces are independent of mass and charge 'cancels' M1

So no deviation A1

*Comments: For (f)(ii), some candidates did not realize that both magnetic force and electric force doubles so they are still equal in magnitude. Leniency in marking saved the day for them.*

(g) (i) packet of energy of electromagnetic radiation B1